
Student Mental Health Wellness Platform Final Year Design Project

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Abstract

NeuroVerse is built to overcome academic stress and socially acceptable barriers to psychological assistance among University students by providing a secure, unified, web-based platform, home to the Psychological Wellness tools for students. The platform features a therapist directory, a digital journal (private for users), and daily mood tracking, built on a scalable and secure MERN stack architecture, including a counselling seminar booking system. It aims to foster open and expressive communication while also ensuring that users feel secure, which is achieved by using an anonymous peer support forum and a light DistilBERT artificial intelligence module for instant text sentiment analysis and distress detection. The platform's robust data privacy measures, such as HTTPS encryption and stateless JWT authentication, ensure it remains accessible and complies with strict data privacy standards, connecting unsafe children with mental health resources.

Keywords: Student Mental Health, MERN Stack, Sentiment Analysis, DistilBERT, Anonymous Peer Support, and Complex Engineering Problem.

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Chapter 1

Introduction

1.1 Project Overview

Nowadays, students face various mental health-related problems, such as anxiety, stress, and loneliness, which significantly affect their academic performance and overall well-being [1, 2]. However, many students do not feel comfortable openly discussing their personal lives. To address this problem, this project proposes a web-based mental health wellness system that allows students to monitor their daily mood, maintain digital journals, and participate anonymously in a supportive community forum. The system also provides counseling support services and easy access to affordable mental health-related resources [2, 3]. The main goal of the platform is to create a safe, accessible, and user-friendly digital environment where students can easily express their feelings and receive suitable mental health support.

1.2 Motivation

In the modern world, mental health issues among students have increased significantly, including academic stress, anxiety, loneliness, emotional distress, and personal struggles [1, 2]. These challenges are further amplified by academic pressure and social expectations. In many cases, students feel uncomfortable and are hesitant to discuss their mental health with friends, teachers, or family members due to social stigma and a lack of understanding. Additionally, existing well-known digital platforms are generally designed for the general public and do not specifically address student mental health [3]. Another major issue is that mental health services are often scattered across many different apps and websites, making it difficult for students to access all necessary resources in a single platform. Therefore, this project is motivated to create **NeuroVerse**, a platform designed to help students cope with anxiety and stress, express their feelings, and gain easy access to mental health resources within a safe, integrated environment [2, 4].

1.3 Objectives

The main objectives of the proposed Student Mental Health Wellness Platform are:

1. To develop a web-based student mental wellness platform that can support at least 100 simultaneous users with an easy and user-friendly interface accessible within 3–4 clicks.
2. To integrate safe mental wellness tools such as daily mood tracking, private digital journaling and emotional trend visualization, while maintaining user information privacy.
3. To establish a peer support forum for students to post and reply anonymously.
4. To offer mental health support resources by incorporating 20+ wellness materials, a therapist directory with at least 10 counseling services, and test the system with at least 20 users to assess usability and effectiveness.

1.4 Methodology

Our proposed Web System development will use the Web System Development Methodology.

Requirement Analysis

Initially, we will determine the primary issues that students have to deal with in terms of mental health. Then we will determine the important features required by students such as mood tracking, journaling, stress measurement, and having mental health resources. This step helps us familiarize with the system requirements and gather the information required in designing the platform.

System Design

Once we finish with the requirement analysis we will design the overall structure of the system. The user interface will be made easy to use and user-friendly to students. The database structure and system architecture will also be designed to specify how various components of the system will interact with each other.

System Development

The development phase will commence after the design phase is done. The frontend will create the user interface, be should be written in HTML, CSS, and JavaScript. The backend will deal with data processing, user authentication, and database.

Testing

Once developed, the system shall be tested to make sure that all parts work out correctly. Functional testing will be conducted to ensure that every feature is functioning correctly. The usability testing will also be carried out so that the platform becomes easy and comfortable to students to use.

Deployment

Lastly, the system will be implemented as a web platform to enable students to access it online. Users will also be able to monitor their mood, to write journals, to access mental after deployment health resources and access mental health support resources.

1.5 Project Outcome

In the current project, we intend to develop NeuroVerse, a mental wellness web application, which will benefit university students who face various challenges in their academic lives [1, 2]. The development of NeuroVerse will be carried out in accordance with the requirements of the MERN stack architecture, which includes MongoDB database, React framework, Express.js server-side environment, and Node.js [5, 6, 7, 8]. One of the main reasons why NeuroVerse needs to be developed is the fact that the university students have to go to multiple websites in order to get the required assistance related to mental wellness, which is inconvenient [3]. Several key features will be offered by NeuroVerse, including anonymous community forum for discussion, daily mood tracker, private journaling feature, and directory of therapy and wellness [9]. Each of these features was included because of the specific problem faced by students, including academic stress, financial stress, emotional stress, and burnout [10, 11]. Overall, the NeuroVerse project does not only satisfy the requirements set for the Final Year Design Project [4] but also is able to contribute to the well-being of students.

1.6 Organization of the Report

The organization of the report is divided into chapters, which will detail the design and development of NeuroVerse mental wellness application system. Chapter 1 introduces the project, giving a brief description of the project, including its background, motivations, objectives, methodology, and anticipated outcomes. Chapter 2 highlights the literature review and related work on the existing mental wellness systems and their shortcomings. Chapter 3 discusses the design and system architecture, such as the system components, database design, and system workflow. Chapter 4 elaborates on the system implementation process, used technology, and other important aspects of the development process. Chapter 5 covers the outcome and performance of the system and provides other impor-

tant findings. The final chapter summarizes the project and makes suggestions for the future.

Chapter 2

Background

2.1 Preliminaries

This section outlines the necessary background information, from which we can obtain the context and rationale for the proposed system. Academic stress, isolation and personal problems have been some of the reasons why we have seen an increase in mental health problems among the students in recent years. Although they may be in need of some help students do not want to seek help because of stigma, lack of education or lack of appropriate tools. With the spread of digital technologies, online mental health programs have become a solution to the above issues. The idea of these services is to create accessible, confidential and user-friendly environments that would allow people to express their emotions, assess their emotional health and seek help. The following are the key concepts that need to be understood in order to design a good and acceptable solution for students such as NeuroVerse.

2.2 Literature Review

The problem of mental health in university students has gained more relevance in recent years with academic pressure, social problems, and personal issues. This has resulted in the creation of multiple digital health solutions such as web and mobile applications to help maintain mental health.

The use of AI in apps such as Woebot [12], Wysa [13] and Youper [14] provides emotional support in a conversational format and simple therapeutic advice. Likewise, the former (MindDoc) [15] concentrates on the mood tracking and symptom monitoring, and the latter (Headspace) [16] prioritizes methods of meditation and stress reduction. Despite the useful services they offer, most of the platforms are not specifically aimed at students, but at the general audience.

Besides applications, recent studies have investigated how artificial intelligence and machine learning could be used to detect and support in mental health. Model-based studies, e.g., BERT and DistilBERT, have demonstrated that text, generated by end users,

e.g., journal entries or social media posts, can be processed to identify certain states of emotion such as stress, anxiety, and depression. These methods demonstrate the promise of automated sentiment analysis in detecting signs of mental health problems.

Regardless of these improvements, there are a number of limitations. There is a plethora of platforms that do not offer anything student-specific, and that do not integrate any other support services or have a combination of multiple services in a single place. There are also limited journaling or tracking features in some of the apps and some are strictly dedicated to meditation. Moreover, lack of personalization and privacy questions decrease the user engagement.

Thus, a significant research deficiency can be seen in the creation of an all-encompassing, student-centered WebAction, integrating mood tracking and journaling, peer support, and professional support (anonymous and accessible). This vacuity rationalizes the creation of NeuroVerse that seeks to offer a more holistic, accessible, and student-friendly mental wellness platform.

2.2.1 Similar Applications

In recent years, various digital platforms have been developed to provide mental health support and wellness tracking. A Table 2.1 is provided showing all the similar application on mental health.

Table 2.1: Similar Mental Health Applications

Application	Type	Primary Function
Woebot [12]	AI chatbot application	Woebot is mainly used as an emotional support system in the form of a conversational AI, through methods founded on Cognitive Behavioral Therapy (CBT) to assist users in controlling their mental health.
Wysa [13]	Mental health application with AI and human coaching	Wysa is a chatbot powered by AI with the option of human assistance through human coaches, offering users both automated and personalized mental health support.
Youper [14]	AI-based therapy application	Youper provides brief therapy sessions powered by AI and enables users to conduct periodic emotional check-ups to monitor their mental health.
MindDoc [15]	Mood tracking application	MindDoc focuses on mood and symptom tracking, provides personalized feedback, and offers self-help courses and exercises to enhance mental health.
Headspace [16]	Meditation and stress management application	Headspace provides guided meditation, mental health care techniques, and tools to assist users in improving focus and achieving personal objectives.

2.2.2 Related Research

Words to Know Emotions

In the recent years, scientists have been investigating the use of artificial intelligence to analyze human expressions by way of writing State-of-the-art language models like BERT and RoBERTa. have been applied to identify depression and anxiety in student generated content [17]. These models concentrate on behavioral data unlike the traditional methods which use behavioral data language structures and mood in writing. These systems examine the manner in which people express themselves and categorize their emotional. state: to be positive, neutral or negative. Research has indicated that such models are very effective and suitable in the diagnosis of mental illnesses. This justifies the usage of lightweight DistilBERT, a model based on APIs, to examine user-created content like journal, posts and posts in forums real time.

The Reason behind being truthful Online

In the recent studies, the question of whether machine learning methods can forecast depression has been studied based on textual data such as diary entries and online posts [18]. The results indicate that people tend to be more open to sharing their genuine feelings online in comparison with the face-to-face interactions. Aspects like the use of word, writing style and posting habits are insightful markers of mental state of a person. This underlines the significance of text-based data as a trustworthy medium of emotional analysis, particularly in an anonymous one.

Application of AI in Mental Health Systems

The other study was dedicated to the introduction of AI-based mental health implementation [19]. Automated models that are a combination of deep learning methods were developed by the researchers as LSTM networks and BERT to assign emotional states such as sadness, fear and anger. These findings showed that these systems have the ability to pick up indicators of mental health problems. Introduces greater efficiency to the conventional screening procedures. This helps in the design of the proposed system, in which a sentiment analysis layer is automated to handle user input recognize the presence of critical emotional states, making it possible to support and intervene.

2.3 Gap Analysis

MindDoc

MindDoc [15] is a popular site that mostly concentrates on mood monitoring among common users. It enables users to track their moods and offers simple journalization. Nevertheless, the platform itself is not targeted at students and does not have any sophisticated functions as anonymous peer support and community interaction. Additionally, its journaling capabilities are also constrained in terms of personalization and more analysis.

Headspace

Headspace [16] is a popular web-based program that puts primary emphasis on meditation and stress management. It offers mindfulness and guided meditations. Nonetheless, it lacks features as mood tracking, anonymous communication and individualized mental health care. Additionally, the site is geared towards a general audience and not necessarily addressing mental health needs of students.

BetterHelp

BetterHelp [20] is an internet-based therapy service, which links the users to professional therapists. It is predominantly applied by people who want to receive structured therapy sessions and counseling services. Despite offering professional assistance, it does not offer

such features as mood tracking, journaling, and peer-based support systems. Moreover, the service is quite expensive and might not be readily available to students.

A detailed feature-by-feature evaluation of these existing systems alongside the proposed application is shown in Table 2.2.

Table 2.2: Comparison of Existing Platforms and Proposed System

Feature	MindDoc [15]	Headspace [16]	BetterHelp [20]	NeuroVerse
Mood Tracking	Yes	None	None	Yes
Journaling	Limited	None	None	Yes
Anonymous Community	None	None	None	Yes
Therapist Access	None	None	Yes	Yes
Target Users	All	All	All	Students

2.4 Summary

A number of research projects have investigated the use of artificial intelligence in mental healthcare systems. One of the studies suggests an AI-driven system in which users communicate with a chat-bot [21] to discuss their mental health issues. The system gathers data by use of standardized tests like anxiety and depression tests, and machine learning models use this data to estimate the need of professional intervention. Then they are then referred to mental health experts to further guidance and treatment. The other research is devoted to applying machine learning algorithms to predict suicide risk detection. It compares survey results concerning mental health symptoms, such as anxiety, depression, and behavior patterns. Various algorithms such as decision trees, logistic, risk levels are classified using regression, and support vector machines. The findings show that these models can be used to effectively identify at-risk individuals, which can be useful in early detection and prevention. Moreover, studies in the field of AI in mental healthcare emphasize the utilization of large datasets and developed models of diagnosis and screening. Chat-bots and other technologies like these can be used. The virtual mental health assistants enhance access and interaction with the user. However, the challenges associated with these systems include limited availability of data, privacy concerns and lack of human interaction. In spite of these drawbacks, AI-based solutions demonstrate high effectiveness, potential and needs more research to be successfully implemented in the real world.

Chapter 3

Project Design

3.1 Requirement Analysis

The analysis of the requirements is a critical part in system design. In this context, it can be used to identify the target users of an automation system - university students and their requirements. Here, we provide a brief summary of the requirements of the new NeuroVerse system based on the literature review and problem identification.

3.1.1 Functional and Non-functional Requirements

To provide efficient mental health services to students, we divided the requirements into two types of requirements: One is functional and another one is non-functional requirements.

Functional Requirements

- **User Registration, login and profile:** This will allow users to register, login and edit profiles. Users will have the option to be anonymous by using display names. Using mood tracker users will be able to keep track of daily mood. It will also allow users to view charts of their moods.
- **Digital Journaling:** Journaling will be provided in a safe and confidential space. There will be an anonymous peer support forum for users built in to the system. It will be an anonymous peer support group where users can express their shock and discuss with other users.
- **Resource Library:** Users can get target setting, tips for improvement, and stress management. Users will be able to find and book therapists or doctors. Users will have access to search and book doctors and therapists.

Non-Functional Requirements

- **Performance:** The web application must be quick to load and support at least 100 users.
- **Usability:** The web app should easy to use. Mood tracking and journaling functions should be available on the homepage.
- **Security and Privacy:** The data is sensitive so we need to ensure users' details are secured and private via authentication and encryption.
- **Availability:** It should be available at all times, enabling users to access it at their own convenience.

3.1.2 Context Diagram

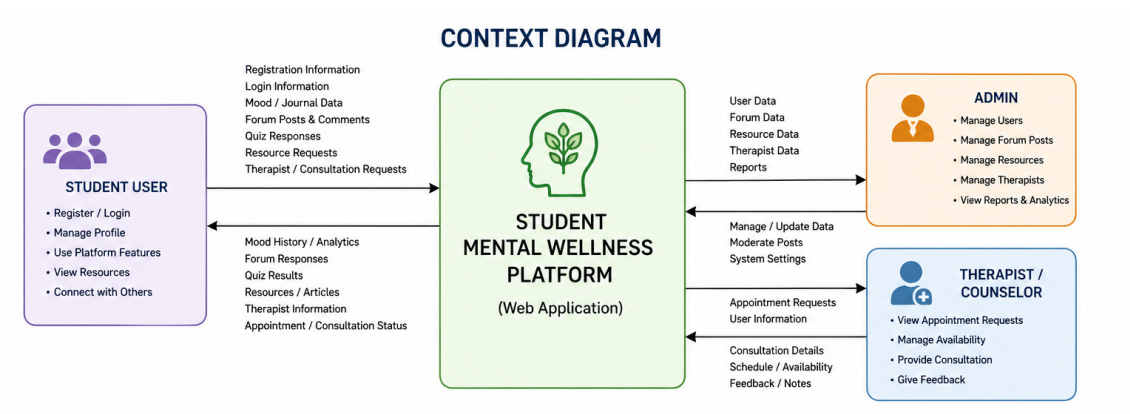


Figure 3.1: Context Diagram of NeuroVerse System

The Figure 3.1 shows the context diagram of the NeuroVerse. The context diagram depicts the interface of the system with external agents such as students, counselors, and administrators. It represents the way the information flows between the user and the platform for various operations.

3.1.3 Data Flow Diagram Level 1

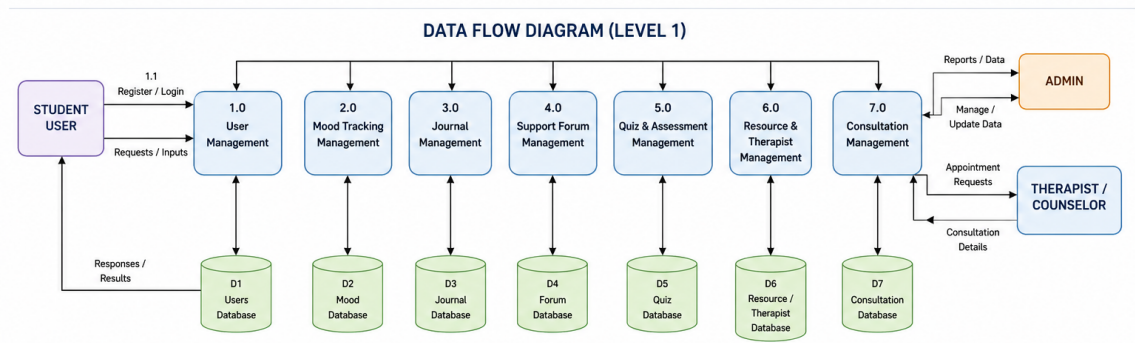


Figure 3.2: Level 1 Data Flow Diagram of NeuroVerse

The Figure 3.2 shows the DFD of the NeuroVerse platform. DFD stands for Data Flow Diagram which describes the information flow through the system. It also shows user inputs, processing, data storage, and output generation process within the system.

3.1.4 UI Design

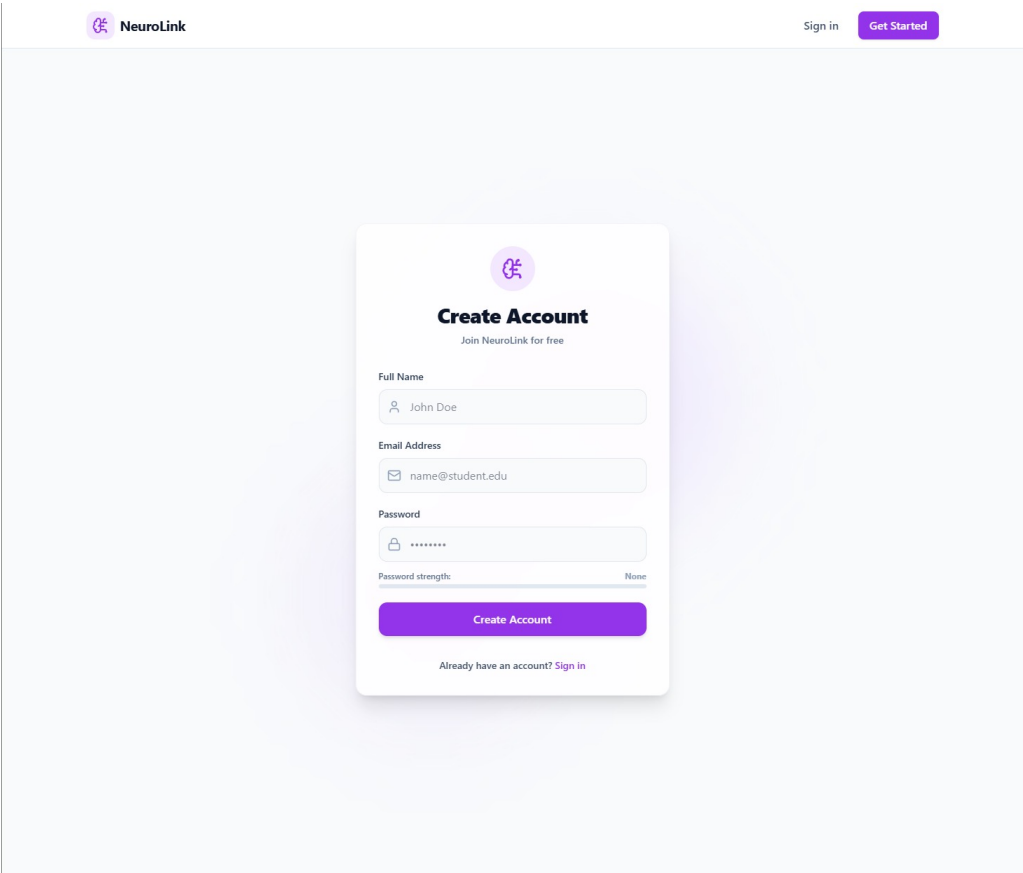
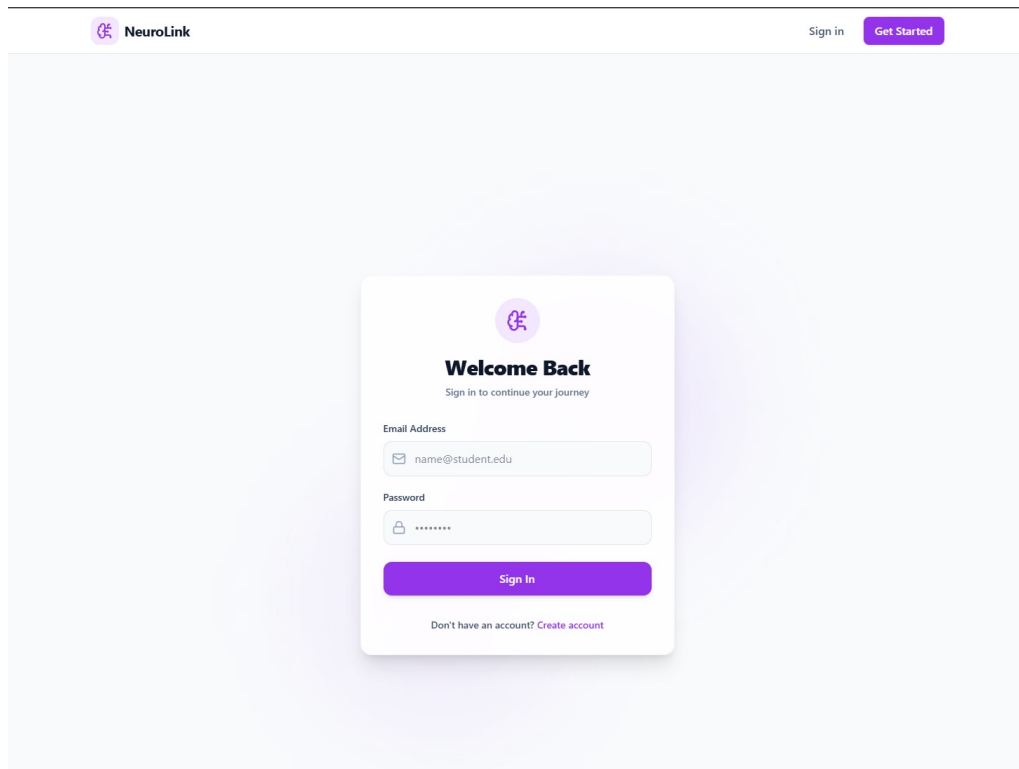


Figure 3.3: NeuroVerse Login Interface Screen

Figure 3.3 is the login screen for NeuroVerse. On this screen users can put in their username and password to get into the NeuroVerse system depending on what their roles.



The image shows a user registration screen for NeuroVerse. At the top left is the NeuroLink logo, and at the top right are links for 'Sign in' and a 'Get Started' button. The main content is a white card with a purple header containing the NeuroVerse logo and the text 'Welcome Back' followed by 'Sign in to continue your journey'. Below this are two input fields: 'Email Address' with the placeholder 'name@student.edu' and 'Password' with masked characters. A purple 'Sign In' button is positioned below the password field. At the bottom of the card, there is a link that says 'Don't have an account? Create account'.

Figure 3.4: NeuroVerse User Registration Screen

Figure 3.4 is the registration screen for the NeuroVerse platform. This is where new users can sign up for NeuroVerse.

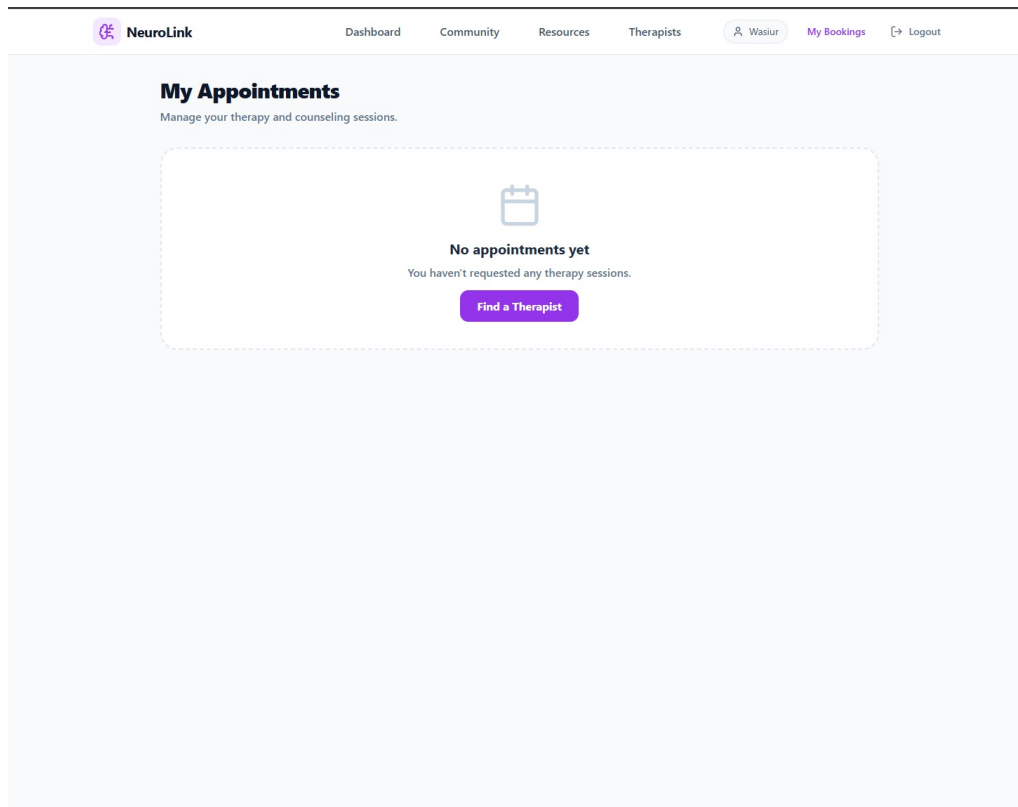


Figure 3.5: Primary Dashboard Interface for NeuroVerse

Figure 3.5 portrays the primary dashboard for the NeuroVerse platform. The dashboard gives easy access to all system features including tracking wellness, scheduling appointments, and receiving notifications.

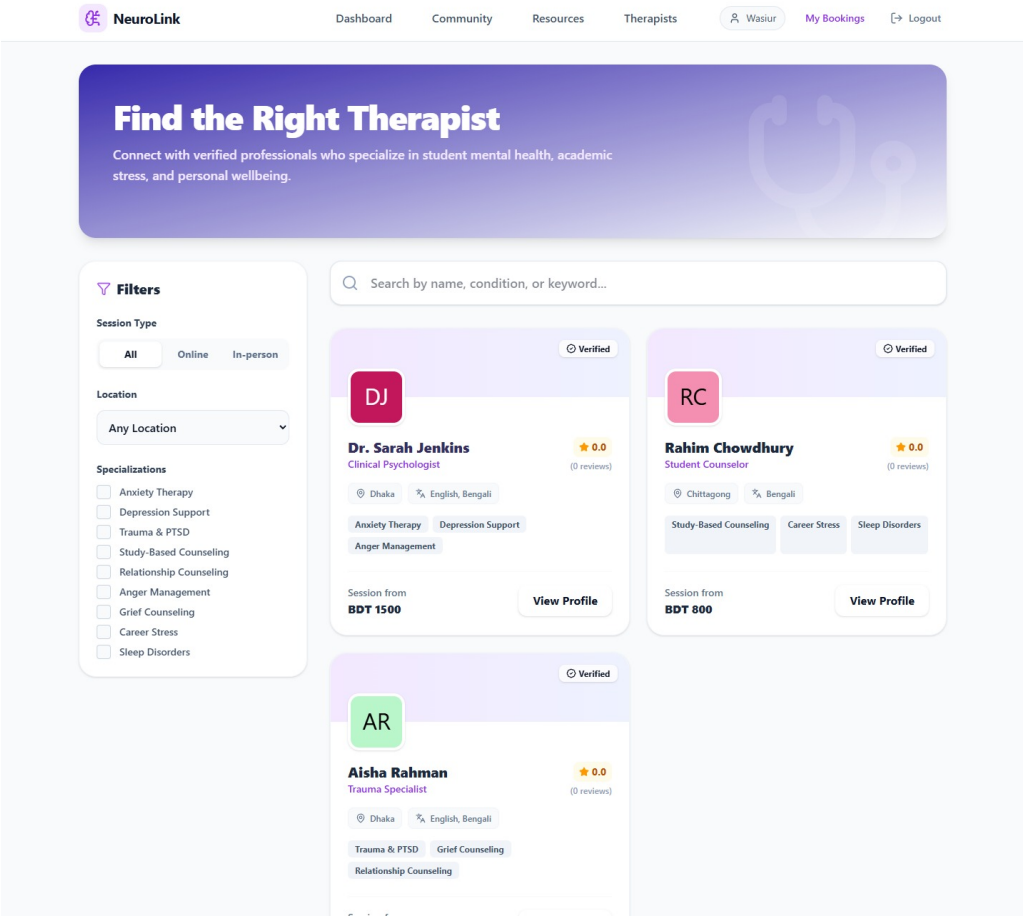


Figure 3.6: Mental Wellness Assessment Interface

Figure 3.6 depicts the Mental Wellness Assessment Interface of the system. Users will be able to submit their assessments and get their feedback about their mental wellness level.

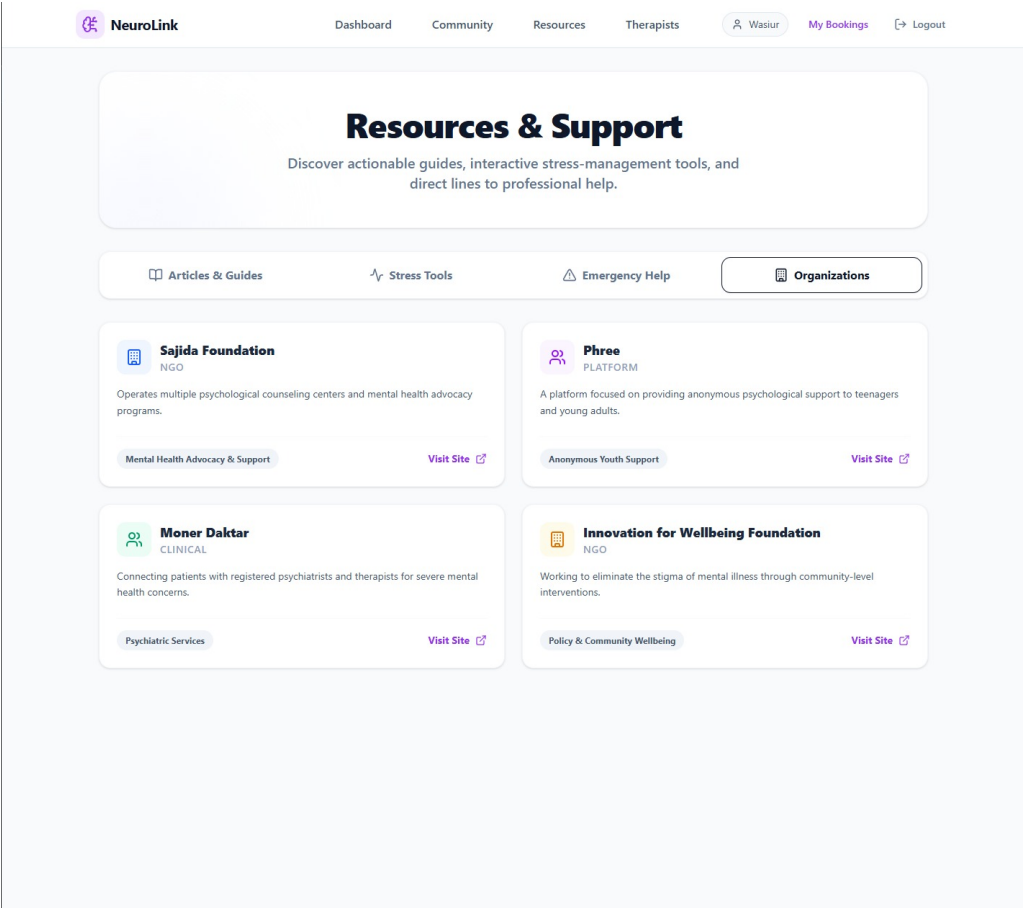


Figure 3.7: Counseling Appointment Management Interface

Figure 3.7 shows the counseling appointment management screen interface. Counseling appointments can be scheduled and monitored using this interface.

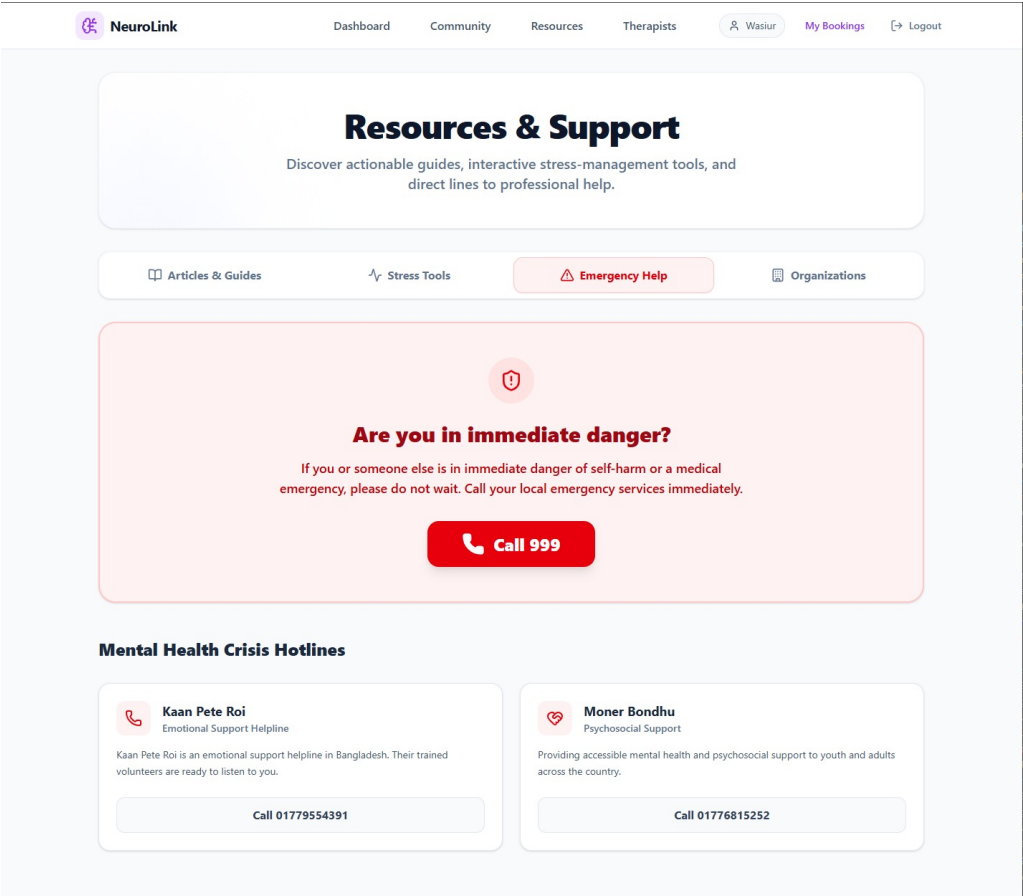


Figure 3.8: Administrator Management Control Screen

Figure 3.8 is the administrator screen for the system. This is where administrators can manage and control activities within the system.

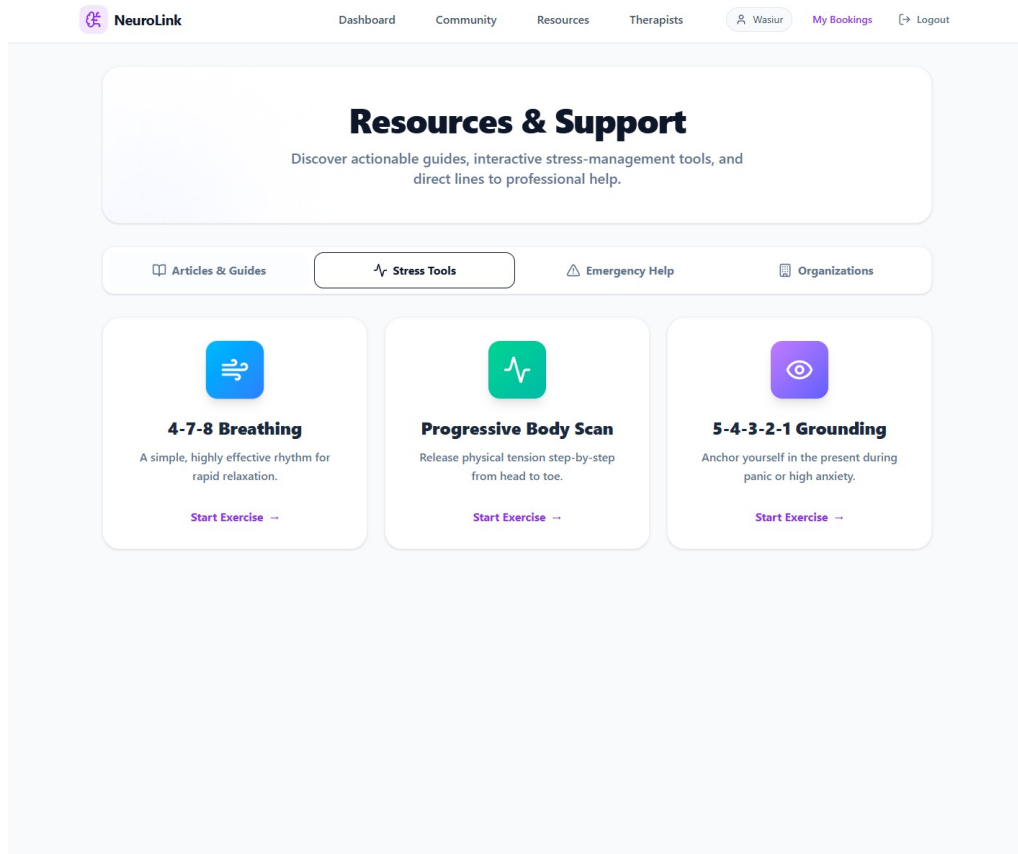


Figure 3.9: Stress Relief Methods Interface

Figure 3.9 is the stress relief methods that are present in the system. These include breathing exercises, meditations, relaxation music, and mindfulness practices, all of which are designed to help the user learn how to relieve stress.

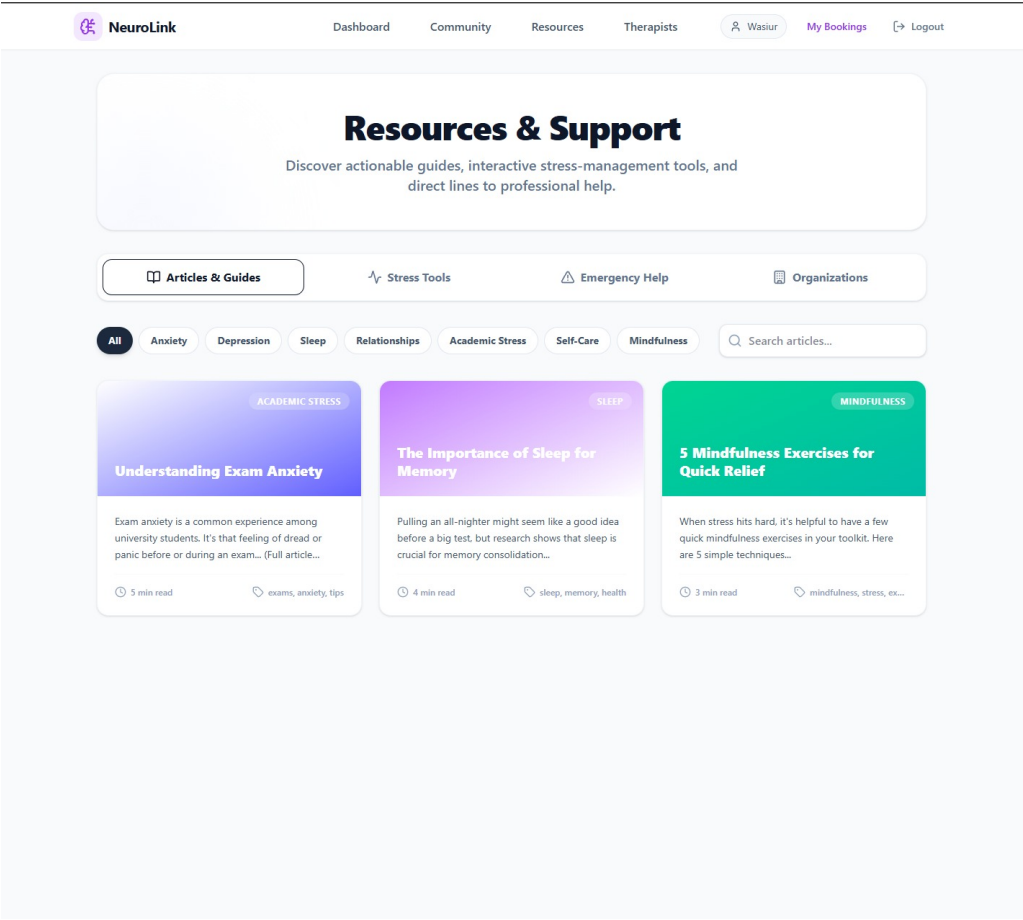


Figure 3.10: NeuroVerse Platform Homepage Interface

Figure 3.10 is the user interface for the homepage of the NeuroVerse platform. The homepage contains all features of the system.

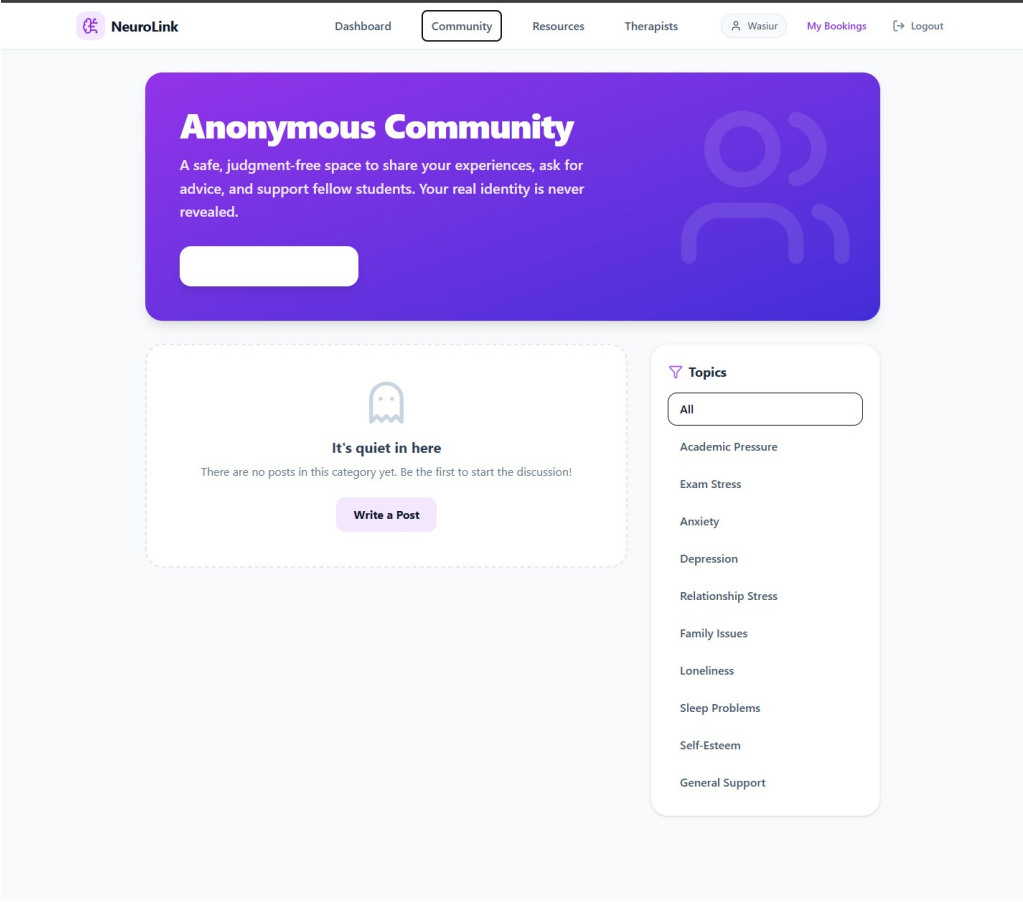


Figure 3.11: Anonymous Mental Health Community Forum

Figure 3.11 shows an anonymous community platform where users can ask any questions or suggestions related to mental health.

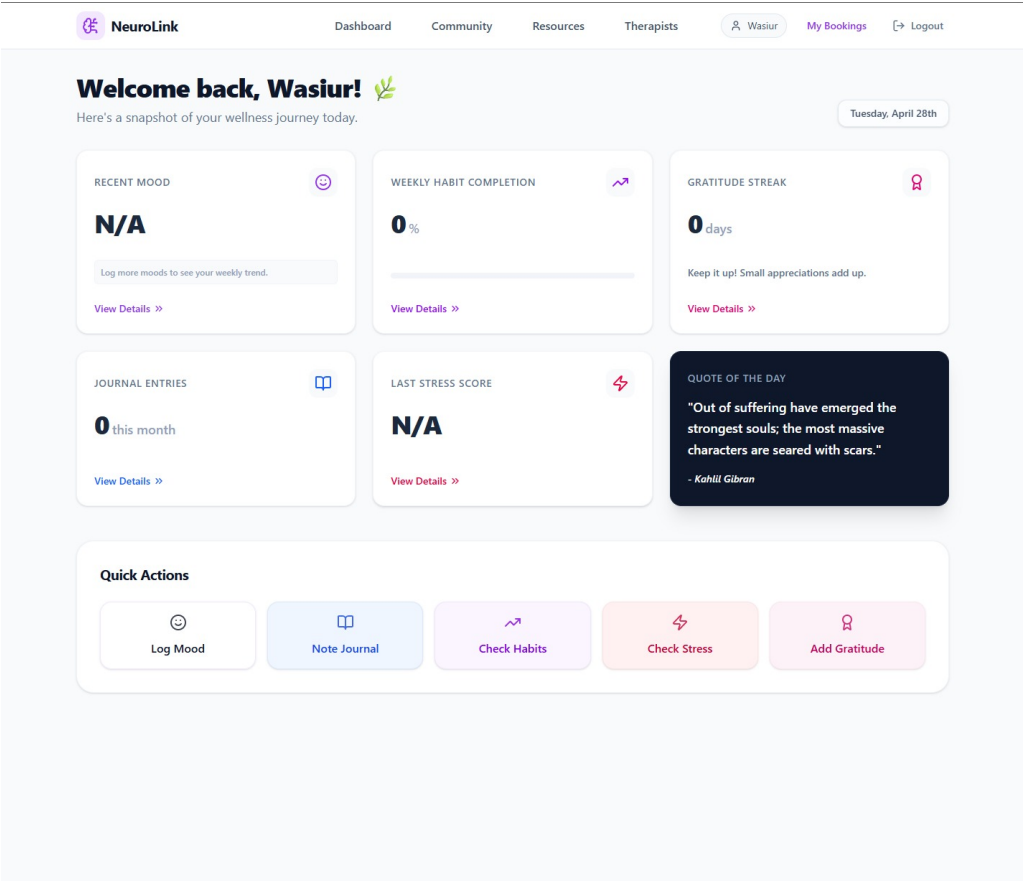


Figure 3.12: Additional System Functionality Interface View

Figure 3.12 shows a further interface view of the NeuroVerse platform illustrating additional system functionality available to user.

3.2 Detailed Methodology and Design

NeuroVerse system: the system will be designed in accordance with Agile Scrum Development Method [22].

This is the most suited development style because the system will be developed in smaller steps in cycles of iterative sprints. This system can be altered according to the feedback of students, counselors, and supervisors as well as according to the Scrum framework, continuous improvement and flexibilities.

Scrum has the development cycle structured around time-boxed sprints where particular features are planned, developed, and reviewed. There is a team of developers that collaborates in every sprint to accomplish the set assigned tasks effectively.

Scrum has several roles assigned in the team:

- **Scrum Master:** The overall duty of the Scrum Master is to oversee the development process and remove obstacles in reaction as well as make sure that the team practices Scrum.
- **Development Team:** The entire team is involved with the process of designing, coding, and testing the system features.
- **Stakeholders/Supervisor:** Review the feedback and directions after every sprint to enhance the system.

Frequent Scrums including sprint planning, daily stand ups, sprint reviews, and retrospectives are useful to keep the team working, monitor the progress and enhance performance.

The project will be divided into sprints which include:

- Sprint 1: Requirement analysis and planning.
- Sprint 2: UI/UX design.
- Sprint 3: Mood tracking, development.
- Sprint 4: Journaling and peer support elements.
- Sprint 5: User feedback and refinement
- Sprint 6: Last testing and deployment.

Reason why Agile Scrum is chosen:

- Supports iterative and incremental development.
- Promote effective teamwork.
- Enables ongoing user feedback and enhancement.
- Assists in coping with changing requirements in a better way.
- Suitable for dynamic Web-based applications.

System Design / Architecture

Web-based three-tier (client, server and database) architecture will be employed in the design of the system.

Main Components:

1. User Interface (Frontend)

This will be a simple and interactive user interface through the web for students to create a student account, log in and record how they are feeling, journal and access strategies.

2. Backend Server

This will handle requests, business logic and the client and the database.

3. Database

It will contain users, moods, journals, forum posts, otherapists and other information

4. Authentication Module

Will be responsible for registering users, authentication, encryption of password and sessions management.

5. Mood Tracker Module

It allows the user to track his/ her mood and progress everyday.

6. Peer Support Module

They will have an anonymous chat with the community.

7. Counselor Support Module

Students will be able to have counselor support or information.

Technology Selection

Our best advice is to use the following:

Frontend:

- Vite
- Tailwind CSS
- React.js [6]

Why: They are used to build modern, responsive and user-friendly UIs.

Backend:

- Node.js [8]

Reason: Node.js is used for creating high performance real-time apps.

Database:

- MongoDB [23]

Reason: Secure databases to store information and user data.

Authentication:

- Firebase Authentication / JWT [24]

Reason: To secure user authentication.

Alternative Solutions Considered

There were alternative solutions to the system developed.

1. Mobile vs Web Application

It was decided that the application will not be a mobile app - this will have to be developed for Android and iOS. This will be more accessible as it will be supported by a browser. Final Choice: Web Application

2. Anonymous Forum vs Direct Chat

Originally direct chat was discussed, but moderating and privacy is a concern. Anonymous peer support provides more anonymity. Anonymous Peer Support Forum

3. Journaling vs Mood Tracker

No tracking was considered, but it offers measurable patterns of moods. Therefore: Journaling + Mood Tracking

Security and Privacy

Security is crucial as data is health related.

Security Measures:

- User Authentication
- Private feature will only be accessible by registered users.
- Encrypted Passwords
- Encrypted passwords will be stored in database.
- Anonymous Posting
- Anonymous posting of forums will be supported.
- Data Privacy Emotional and personal information will be confidential.
- Secure Communication
- We will communicate over HTTPS.

3.3 Project Plan

NeuroVerse is developed in a structured and disciplined way with the system divided into a number of phases over a three trimester time period to maximise time. We are using the Agile scrum development method [22] which includes iterations with feedback. There are five stages in the life cycle namely requirements analysis, system design, implementation and development, and system deployment.

Project Phases

Phase 1: Requirement Analysis and Planning

This is where the project goals and requirements are outlined. Functional and non-functional requirements are collected, and a literature survey is undertaken. Relevant platforms are evaluated to identify their pros and cons.

Phase 2: System Design

The system design is developed in line with the requirements. Further model diagrams such as context diagrams and data flow diagrams are developed. The system is also designed with the UI/UX design.

Phase 3: System Development

The system is developed in this phase using the MERN stack. Development is in two parts:

- **Frontend Development:** Creating the user interface with React, including mood logging, journal entries and anonymous forums.
- **Backend Development:** Authentication, database integration, API development and security and privacy.

Phase 4: Testing and Evaluation

Functionality of all integrated system parts is tested. Functional testing and usability testing. We gather data from at least 7 students to enhance system performance and functionality.

Phase 5: Deployment and Documentation The system is made available on a web hosting service. The documentation of the system gradually evolves to report the development process and results.

Project Schedule and Milestones

Here's the Table 3.1 for estimated project timetable and milestones to be completed within three academic trimesters (30-32 week).

Table 3.1: Project Schedule and Milestones

No.	Activities	Time Span	Deliverable
Phase 1	Review of literature, identify the gaps, aim of the project, requirement gathering	Weeks 1 - 6	Project Proposal, SRS Document
Phase 2	UI/UX working on design, context diagram, database structure, entity relationship diagram (ERD)	Weeks 7 - 10	System Design Document, UI/UX
Phase 3	Platform functionality (web front-end, back-end API)	Weeks 11-24	Working Platform Prototype
Phase 4	Testing, evaluation and validation	Weeks 25-28	Report on Testings, New System
Phase 5	Bug fixing, deployment, report writing, and defense preparation	Weeks 29-31	Deployed System, Final FYDP Report

3.4 Task Allocation

To successfully accomplish NeuroVerse, we have to allocate tasks to team members carefully based on skills and needs in the project. Different tasks are being provided to the members considering their knowledge in the following Table 3.2.

Table 3.2: Distribution of Task in the Team

Assigned Students	Stu-ID	Activities
Muhaiminul Amin Shafin	011221456	Project Management & Backend Development. Handling the system's design, API and database creation, project planning and management.
Md. Annan	011222128	AI and Sentiment Analysis. Developed the artificial intelligence module that takes users' text data (journal, post) for understanding the emotional state of the user.
Md Asif Sarkar	0112230037	UI/UX & Frontend Development. UI design (Figma) and Frontend Development (HTML, CSS, and React.js).
Mirza Samawatun Nur Astha	0112230887	Backend & Frontend Development and Feature Development. Implemented key features including mood tracking, journal entries.
Wasiur Rahman Sakib	0112230747	Firebase & Web Deployment. Manages authentication, real time database and code testing.

3.5 Summary

This chapter described the strategy used in designing and building the proposed Student Mental Wellness Platform NeuroVerse. The requirements were analysed, considering both functional and non-functional requirements. System and communication structure was represented in the diagrams.

The interaction design: ease of student navigation. The methodology chapter contained information about the system's Agile [22] approach and architecture. Comparative studies of designs were carried out.

Also, the project plan and task allocation provided a structured process of system development and workload among the developers. Overall, this chapter provided a roadmap for the development process towards an efficient and secure system for students.

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

5.1 Compliance with the Standards

In the development of the NeuroVerse Student Mental Health Wellness Platform we make sure to follow industry standards for security, scalability and compatibility. The platform uses web practices and secure communication protocols to support a reliable infrastructure for university students.

5.1.1 Software Standards

1. Secure Web Communication

- **Description:** The system uses HTTPS (Hypertext Transfer Protocol Secure) with TLS (Transport Layer Security) for all data transfers as defined by the Internet Engineering Task Force (IETF) standards [25, 26].
- **Alternates Considered:** HTTP (communication).
- **Pros of Alternates:** Easier to set up for development and uses less server power.
- **Cons of Alternates:** HTTP sends data in plain text, making sensitive information vulnerable to interception.
- **Rationale for Selection:** NeuroVerse handles highly sensitive mental health information, so it must ensure absolute data privacy. HTTPS encrypts data during transit, maintaining confidentiality and user trust.

2. Secure Authentication

- **Description:** The platform uses Firebase Authentication and JSON Web Tokens (JWT) for secure user session management in accordance with RFC 7519 standards [27].
- **Alternates Considered:** Server-Side Session Cookies.

- **Pros of Alternates:** Easy to implement on monolithic applications and handles session invalidation seamlessly.
- **Cons of Alternates:** Requires the server to store session states in memory or a database, which can reduce backend scalability.
- **Rationale for Selection:** The MERN stack (Node.js, Express, React) works best with stateless authentication. JWTs allow the backend to quickly verify user identities cryptographically, keeping the system fast, stateless, and secure.

3. Web Frontend Standards

- **Description:** The frontend architecture follows W3C web standards using React.js, Vite, and Tailwind CSS [28].
- **Alternates Considered:** Native mobile application development (e.g., Java/Kotlin for Android, Swift for iOS).
- **Pros of Alternates:** Better optimization and native use of mobile device hardware.
- **Cons of Alternates:** Requires maintaining separate codebases, resulting in significantly longer development times.
- **Rationale for Selection:** A web-based application provides immediate cross-platform access across any browser-enabled device. This allows the NeuroVerse platform to be fully responsive and instantly functional for all students without installation barriers.

4. Data Serialization Standards

- **Description:** JSON (JavaScript Object Notation) is used for exchanging data between the frontend and backend APIs, as well as database storage via MongoDB [29, 30].
- **Alternates Considered:** XML (eXtensible Markup Language).
- **Pros of Alternates:** Highly structured with rigid schema validation rules.
- **Cons of Alternates:** Verbose syntax with heavy overhead, making it significantly slower to parse.
- **Rationale for Selection:** JSON is native to JavaScript and document databases. Utilizing JSON allows seamless, high-performance communication between the React frontend, Node.js server, and MongoDB, increasing overall system efficiency.

5.1.2 Hardware Standards

The NeuroVerse Student Mental Health Wellness platform is developed hardware standard and cloud computing guidelines to ensure that reliable performance ,easily access and also

system availability. These are essentially no particular hardware certain to use using this sort of platform since it is mostly web based. Can be easily used to access from any devices commonly use like smart phone, laptop, computer and tablets.

1. **Client Device Standards** The platform can be used by any modern web browser such as Google Chrome [31], Mozilla Firefox [32], and Microsoft Edge [33]. It can be easily accessed by students on tablets, laptops, and mobile devices.

Reason for selection: The client device Standard device has been selected as it allows users to easily access the platform without any additional equipment. This is more accessible and also helps to cut costs [34].

2. **Cloud Server Infrastructure Standards** The platform built on cloud servers is what helps to provide us with computing power, data processing services and network services. Cloud Server Infrastructure Standards is expected to line up the help with it.

Reason for selection: Cloud servers provide us with several benefits such as scalability, high availability, speed in terms of recovery support along with cost and less expensive, helps in the creation of modern web applications [35].

3. **Network Hardware Standards** The platform uses standard internet networking technologies mainly based on the TCP/IP protocols. Common networking equipment would include broadband, wireless access points, switches and routers.

Reason for selection: It is easier to access the Internet-based networking without any special networking equipment and is much more flexible and accessible [36].

5.1.3 Communication Standards

To ensure that secure data, efficiency and reliable communication throughout the NeuroVerse platform, this system architecture by several communication standards. These standards mainly help to maintain the data security, smooth client-server interaction, and reliable cloud communication throughout the system.

- **Hypertext Transfer Protocol Secure (HTTPS):**

All the communication between the fronted work by React and the backend work by Node.js is secured using the HTTPS encryption. This protocol protects sensitive data from unauthorized access or interception [37].

- **JSON Web Token (JWT) Authentication:**

All the communication between the fronted work by React and the backend work by Node.js is secured using the HTTPS encryption. This protocol protects sensitive data from unauthorized access or interception [24].

- **HTTP-Only Cookie Communication:**

Authentication tokens are store and transmitted using HTTP-only the reduce the risk of Cross-Site Scripting (XSS) attacks [38].

- **RESTful API Communication:**

platform uses mainly connect RESTful APIs with Express.js middleware to manage communication between the client in fronted and server securely and efficiently in backend [39] [7].

- **Secure Cloud Communication:**

MongoDB Atlas and the cloud infrastructure using encrypted connections and secure networking technologies, the safe and protected communication between backend and the database [23].

5.2 Design Constraints

Only mention the constraints that are related to the design of your project. This list is not complete.

5.2.1 Economic Constraint

5.2.2 Environmental Constraint

5.2.3 Ethical Constraint

5.2.4 Health and Safety Constraint

5.2.5 Social Constraint

5.2.6 Political Constraint

5.2.7 Sustainability

5.3 Cost Analysis

The proposed system, called NeuroVerse: Student Mental Health Wellness Platform, is a web application that is software-based and based on the use of modern open-source technologies. The cost of this development is relatively low because the project uses free open frameworks and cloud-based services [4]. The system does not need the purchase of the costly infrastructure hardware which will make it to be a cost effective system for academic and future implementation. A system's frontend is built with React.js, Vite, and Tailwind CSS [6]. Both of these technologies are free of charge for licensing and free of licensing fees. Likewise, the backend is crafted in Node.js, another free and popular language for fast and scalable web application development [7, 8]. Owing to its flexibility and power to store journal entries, mood details and random forum posts given by users,

MongoDB will be used for Database Management [5, 40]. It also utilizes the Firebase Authentication / JWT for safe logins and account management capabilities for users [24, 27]. Firebase provides its services in a free tier that are appropriate for development and testing needs. The majority of development activities will be done with individual laptops and connectivity with the Internet, of the development team. In this way you do not have to worry about any additional hardware to get the system. The anticipated costs include domain registration, deployment costs, Internet usage, and testing. For future expansion with a higher user base, further expenses could be needed for cloud server scaling, sophisticated database storage capabilities, maintenance, and increased security measures [23].

A comprehensive breakdown of these development and operational expenses is outlined in Table 5.1.

Table 5.1: Estimated Development Cost

Item	Estimated Cost (BDT)
Domain Registration	1,500
Cloud Hosting / Deployment	2,500
Internet and Communication	3,000
Testing and Miscellaneous Expenses	1,500
React.js, Vite, Tailwind CSS	Free / Open Source
Node.js	Free / Open Source
MongoDB Database Services	Free Tier
Firebase Authentication / JWT	Free Tier
Documentation and Printing	1,000
Total Estimated Cost	9,500 BDT (Approx.)

5.4 Complex Engineering Problem

The project, "NeuroVerse - Student Mental Wellness Platform", tackles a societal and real-life challenge pertaining to student mental health and emotional wellbeing [9]. University students are frequently stressed, anxious, lonely, depressed and under emotional pressure, but many may be reluctant to seek help for those reasons due to stigma, lack of knowledge, and the lack of professional help systems available [9].

The project aims to provide a platform for students to anonymously access mental health resources, communicate with each other, access self-help tools and engage with mental health consultants. It has several features such as tracking mood, anonymous community support, therapist directory, stress management tools, mental health resources, consultant booking systems, and wellness dashboards all within one system.

The project is classified as a Complex Engineering Problem (CEP), where various technical, social, security and usability issues need to be addressed simultaneously. Incorporat-

ing frontend technologies, backend services, secure authentication, database management, cloud implementation, role-based access control, and real-time user interaction is essential for the successful integration into the system [5, 6, 7, 8]. Moreover, the platform handles emotionally and psychologically sensitive data, creating significant privacy and security issues.

The complexity of the project is also due to:

- Providing for the safe and confidential exchange of ideas, and
- Maintaining user privacy,
- Managing different stakeholders,
- Ensuring system scalability,
- Combining various modules in a single platform,
- Adopting a user-friendly and pupil-friendly interface,
- Moderating community content that can be harmful or inappropriate.

The project is implemented in MERN stack architecture which is scalable, deployed in cloud environment [5, 6, 7, 8]. To make the system maintainable and ensure the security of user data protection, JWT authentication and a modular front end architecture are used [24].

Hence, the proposed system shows the features of a Complex Engineering Problem based on multidisciplinary engineering knowledge, dependency of stakeholders, social impact, secure system design, and the development of an integrated software system in large scale.

5.4.1 Complex Problem Solving

The proposed project is mapped with the characteristics of problem solving based on the standard engineering problem solving classification. For each mapping criteria, a detailed rationale is provided in the subsections below as summarized in Table 5.2.

Table 5.2: Mapping with Complex Problem Solving

Problem Solving Criteria	Mapping	Rationale
P1- Depth of Knowledge Required	✓	Must have knowledge in MERN stack development, database management, cloud computing, cybersecurity, authentication systems, UI/UX design, and software architecture.
P2- Range of Conflicting Requirements	✓	Anonymity, data security, performance, usability, moderation and accessibility are conflicting requirements that must be met at the same time.
P3- Depth of Analysis Required	✓	Detailed analysis is required for mental wellness workflows, user interaction systems, database relationships, dashboard analytics, and role based functionalities.
P4- Familiarity of Issues	–	Limited attention is usually given to mental health platforms which are anonymous in nature & the project entails unfamiliar social and ethical challenges.
P5- Extent of Applicable Codes	✓	The project is designed to adhere to secure implementation of authentication practices, data privacy principles, ethical content moderation and secure cloud implementation.
P6- Level of Stakeholder Involvement	✓	The system includes several stakeholders such as students, therapists, consultants, moderators and administrators.
P7- Interdependence	✓	All the modules, like mood tracking, anonymous support, booking of the consultant, dashboards, and resources are interconnected and dependent on each other in a system.

P1: Depth of Knowledge Required

The project requires extensive expertise in the MERN stack (MongoDB, Express, React, and Node) for web development, cloud database management with MongoDB Atlas, JWT authentication, frontend component architecture, and secure API development [5, 6, 7, 8]. Deep knowledge regarding user-centered UI/UX design and digital mental wellness frameworks is also mandatory to complete the implementation [9].

P2: Range of Conflicting Requirements

The system should anonymize user profiles while at the same time avoiding misuse of the anonymous support forum. It must balance strict data security protocols during the handling of sensitive emotional information with an open, completely friction-free user interface to ensure students in distress are never discouraged from accessing help.

P3: Depth of Analysis Required

Creating scalable cloud database schemas, securely tracking multi-tier user activities, developing data-driven wellness dashboards, implementing a multi-role consultant booking calendar, and establishing community moderation systems all require extensive engineering analysis. The platform must also thoroughly analyze student accessibility constraints.

P4: Familiarity of Issues

The proposed platform goes completely beyond traditional management information systems, taking a unique mental health and emotional wellbeing lens. This introduces less familiar engineering and design challenges regarding human emotion tracking, ethical automated text monitoring, social stigma mitigation, and digital psychological safety [9].

P5: Extent of Applicable Codes

To ensure the safe and ethical handling of highly sensitive data, the system strictly implements modern software engineering practices and data privacy codes. This includes secure JWT authentication tokens, protected HTTP-only tracking cookies, encrypted cloud communication layers, and robust content moderation policies [24].

P6: Extent of Stakeholder Involvement

This project heavily involves multiple parties with deeply distinct roles and system access permissions. The engineering design must satisfy students seeking anonymous support, consultants offering counseling services, therapists arranging professional appointments, and platform administrators managing overall backend operations.

P7: Inter-dependence

The modules built across the platform are deeply interdependent. Wellness tracking inputs directly generate data analytics for the user dashboards; role-based session authentication is critical to facilitate the consultant booking workflows; and automated community moderation directly synchronizes with the core database architecture to enable safe, real-time anonymous forum interactions.

5.4.2 Engineering Activities

The creation of the NeuroLink platform involves tackling intricate software engineering challenges, such as establishing its scalability, managing secure user interactions, and processing transactions and payments. In this section, specific attributes of our Student Mental Wellness Platform are aligned to core engineering activities in Table 5.3.

Table 5.3: Mapping of Project Features to Engineering Activities

Project Feature	Core Engineering Activity	Mapping of Project Features to Engineering Activities
Anonymous Support Forum, Content Moderation System	EA1: Algorithm Design & Data Structures	Designing algorithms and suitable data structures for anonymous support forum operations and content moderation system.
Therapist Directory & Course Module	EA2: Database Design & System Architecture	How to design the MongoDB schema [40] for the relationship between users, consultants, booking slots and course enrollments.
User Auth, Journal, & Mood Tracking	EA3: Security & Privacy Engineering	Secure authentication [27], encrypting private journal entries, and logic for the dashboard for habit/mood tracking.
Consultant Booking & E-Commerce	EA4: Secure API & Full-Stack Integration	Securing API [41] & Full-Stack Integration for booking requests (Exam Stress Counseling) and course transactions (500 BDT for Anger Management).
Stress Quiz & Progress Dashboard	EA5: Testing, Verification & Validation	Test cases for Quiz scoring algorithms and validation of self improvement data visualization accuracy for Stress Quiz & Progress Dashboard.

EA1: Algorithm Design and Data Structures

Mapped Features: Anonymous Support Forum, Content Moderation System.

Rationale: Algorithmic planning is needed for the creation of an anonymous community where students would be able to share sensitive feelings such as: “I feel very lonely these days” [42]. In an engineering sense, this means to design efficient data structures to support threaded comments which enable users to observe every comment on a page, to support reactions, and to do filtering of comments by category (Anxiety, Academic Pressure). In addition, it is necessary to manipulate an automated or semi-automated moderation platform to ensure adherence to community rules and mark content as “high risk” without jeopardising anonymity [43].

EA2: Database Design and System Architecture

Mapped Features: Consultant Profile Management, Course Enrollment Tracking.

Rationale: It is a data-driven platform. A scalable schema (MongoDB or other NoSQL) [40] needs to be designed that will be able to connect up multiple disjoint entities. For instance, the system should correlate Users to the purchased Courses, such as the Overcoming Anxiety, 500 BDT course, and should also record their movie progress. At the same time, it should be able to handle “Consultant Profiles” such as Dr. Suman or Dr. Hassan, and keep a record of their particular rates of service, as well as the availability of their scheduling. This needs to be done on an architectural level, which means avoiding

anomalies in the database.

EA3: Engineering for Security and Privacy

Mapped Features: User Authentication, Personal Journal, Mood Tracking, Daily Gratitude Log.

Why it isn't sufficient: NeuroLink requires the ability to handle sensitive mental health information, and conventional CRUD (Create, Read, Update, Delete) operations are insufficient [43]. The engineering challenge here is data security. A strong User Authentication system (Signup/Login) is mandatory requiring the use of JWT (JSON Web Tokens) or OAuth [27]. Moreover, private aspects of the system, such as the Personal Journal and Mood Tracker require that students' entries are protected and perhaps even encrypted at rest so that even a system administrator cannot easily read a student's private thoughts.

EA4: Secure API and Full-Stack Integration

Mapped Features: Consultant Booking System, Course Enrollment System, Emergency Help Section.

Rationale: Engineering secure RESTful APIs [41] is important to connect the frontend interfaces (such as Course Details Page or Consultant Profile) to the backend database. If the user registers for "Study Motivation Session" with 1200 BDT or tries to register for a course, the system should process the request correctly, update the user's dashboard and inform the consultant. This activity should include managing state on the front end (such as React), processing asynchronous backend requests (such as Node.js/Express), and keeping the system from failing to respond to user requests that are made concurrently.

EA5: System Testing and Logic Validation

Mapped Features: Stress Quiz, Progress Dashboard, and Habit Tracker.

Rationale: A mental health platform should not be unreliable. The Stress Quiz needs to be validated for logic in order to ensure that the user's inputs are calculated correctly to give the user the correct mental health assessment score. Further, the Progress Dashboard, which combines data from the sleep/exercise Habit Tracker and Mood Tracker, needs to be robustly unit tested. The engineering rationale here is that if the self-help toolkit mirrors the student's progress, in an accurate and bug-free way, it will truly support the student's self-improvement process.

5.5 Summary

This chapter presented the key generational requirements, constraints, cost-budgets and intricate engineering factors that determine the NeuroVerse platform. The platform does so using best practices employed in web design and architecture, including secure cloud data networking [23] , secure implementation of JWT [27] with a stateless approach, and encrypted HTTPS [37] data transmission, ensuring data privacy and absolute confidentiality for vulnerable groups facing mental health issues [2].

The approach to the structural blueprint is built on a cost-effective and highly efficient open-source tools that use the MERN stack [6, 7, 8, 40] and requires limited overhead of initial deployment to provide maximum usability. Finally, by laying out the platform’s various needs over common indices of problem solving, mapping the needs reveals that providing secure, real-time and anonymous digital interventions to help reduce student mental health distress is a highly interdependent Complex Engineering Problem (CEP) that demands a delicate balancing act of algorithms and architecture.

Chapter 6

Conclusion

This chapter highlights the results and impacts of the entire NeuroVerse: Student Mental Health Wellness Platform project. It also explains the limitations of the present system and provides suggestions for future improvements.

6.1 Summary

Mental health becomes increasingly relevant today among students of modern educational establishments [1, 2]. Numerous students are stressed, experience anxiety and depression, and face emotional strain; however, they hesitate to turn to professionals to address the issue. The “NeuroVerse: Student Mental Health Wellness Platform” project was initiated to provide efficient web solutions for addressing mental health needs of students. The primary aim of the project was to develop an online platform providing students with means for tracking their mental health status, getting appropriate self-help information, contacting professional counselors, and getting relevant guidance. During the development, software engineering approaches, system analysis methods [4], and latest technologies were applied in order to provide reliable, user-friendly, and maintainable systems [3, 34]. The developed system incorporates numerous functions such as user authentication [24, 27], provision of mental well being assessment facilities, counselors’ contact and guidance options, as well as managerial functions. Functional and non-functional requirements were analyzed during the process of software development [4]. In addition, the project illustrated the potential of technology in making positive contributions towards social welfare through creation of a conducive digital space for students. NeuroVerse serves as a solid base for fostering awareness concerning mental wellbeing as well as provision of psychological assistance services [9].

6.2 Limitation

Despite having been able to meet the initial goals set by the team, there are still quite a few drawbacks to the system currently. Firstly, the system relies almost solely on

counseling help provided manually and lacks Artificial Intelligence or other systems that provide recommendations for personal mental health evaluation. Secondly, the platform is currently accessible only through a web-based interface [31, 32] and does not offer any separate mobile apps for Android or iOS devices yet. Moreover, the facility to offer real-time emergency services is quite limited, and there is no provision within the current system to offer any means for instant crisis intervention. Likewise, there are no video-based counseling facilities or live chat options currently available. Last but not least, it is not feasible within the time-frame and cost constraints to perform large scale testing among many students. One of the limitations associated with the system is the absence of advanced analytics features.

6.3 Future Work

A number of features that can be enhanced in subsequent versions of NeuroVerse to enhance its efficiency and efficacy are: The incorporation of technologies related to Artificial Intelligence (AI) and Machine Learning to offer personalized advice for the users along with sentiment and risk analyses [10, 11, 19, 21]. Additionally, a specific mobile app for the iOS and Android operating systems can also be developed to offer greater user convenience. Some of the features that could be included in later versions of NeuroVerse include emergency support services, immediate alerts, hot-line integration, secure video counseling sessions between professional counselors and users, etc. Advanced analytics and reporting tools could also be incorporated into the system for better understanding by the administrators and counselors regarding student well-being trends and behavior. Furthermore, additional languages can also be offered to the users of NeuroVerse as a way of making it even more inclusive [18]. Improved privacy and security protocols through biometric authentication and advanced data encryption techniques could also be incorporated [23].

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Contribution

This section presents the individual contributions of each team member in the development of the NeuroVerse: Student Mental Health Wellness Platform.

1. Muhaiminul Amin Shafin

- Section 1.4: Methodology
- Section 2.3: Gap Analysis
- Section 3.1.2: Context Diagram
- Section 3.1.3: Data Flow Diagram
- Section 3.1.4: UI Design
- Section 5.4 Complex Engineering Problem
- Section 5.4.1 Complex Problem Solving

2. Md. Annan

- Section 1.2: Motivation
- Section 2.2.1: Similar Applications
- Chapter 3: Project Plan
- Chapter 5.1.2 Hardware Standards
- Chapter 5.1.3 Communication Standards

3. Mirza Samawatun Nur Astha

- Section 1.3: Objectives
- Section 1.6: Organization of the Report
- Section 2.1: Preliminaries
- Section 2.2: Literature Review
- Section 3.2: Detailed Methodology and Design
- Section 3.5: Summary
- Section 5.3 Cost Analysis
- Section 5.5 Summary

4. Md Asif Sarkar

- Section 1.1: Overview
- Section 2.4: Summary
- Section 3.4: Task Allocation
- Section 5.1 Compliance with the Standards
- Section 5.1.1 Software Standards

5. Wasiur Rahman Sakib

- Section 1.5: Project Outcome
- Section 2.2.2: Related Research
- Section 3.1: Requirement Analysis
- Section 3.1.1: Functional and Non-Functional Requirements
- Section 5.4.2 Engineering Activities